



Multi-layer feature fusion for high-accuracy solid waste classification using a hybrid deep learning model

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Abstract

Solid waste management is crucial for environmental sustainability, and effective waste classification systems are vital due to the escalating volume of waste. This study proposes an innovative deep learning model that integrates EfficientNetB0, InceptionV3, and the HyperColumn technique for waste image classification. The model aims to learn multi-resolution features more effectively by combining the strengths of these architectures. EfficientNetB0 provides high performance with fewer parameters, while InceptionV3 captures multi-scale features using various filters. The HyperColumn technique fuses feature maps from different layers of both models, enabling the extraction of both general and detailed features. The XGBClassifier is employed in the classification phase to optimize final performance. Experimental results on the Household_Garbage and TrashNet datasets demonstrate the model's superiority, achieving 99.87% accuracy, 99.77% precision, 99.81% recall, 99.79% F1-Score, 99.99% specificity, and 99.90% AUC on the Household_Garbage dataset, and 99.40% accuracy, 99.49% precision, 99.54% recall, 99.51% F1-Score, 99.88% specificity, and 99.71% AUC on the TrashNet dataset. These results highlight the model's accuracy and balance, positioning it as an important step towards sustainable waste management through automated and efficient solutions. The code and data are available at <https://github.com/GaffariCelik/SolidWasteClassification>.

Keywords Solid waste · Deep learning · EfficientNetB0 · InceptionV3 · HyperColumn · XGBClassifier

1 Introduction

Inadequate management of solid waste is considered one of the most important causes of environmental pollution at the global level. The increase in single-use products in particular, has caused a significant increase in the amount of waste in recent years. According to Eurostat's 2020 data, the amount of waste per capita in the European Union countries is 4.8 tons [1], while according to the estimates of the World Bank (WB), this amount is expected to exceed 3 billion tons per year by 2050 [2]. According to the same report, only 13.5% of global waste is recycled, while approximately 33% is directly disposed of into the environment without any classification [2]. This situation causes different types of waste to spread uncontrolledly into nature; plastic waste in particular stands out as one of the biggest threats due to its long-term environmental effects [3, 4].

According to research, approximately 8.3 billion tons of plastic were produced worldwide between 1950 and 2015, and approximately 6.3 billion tons of this became waste. However, only 9% of this waste was recycled, 12% was incinerated, and the remaining 79% was dumped in landfills or left in nature [5]. While the time it takes for plastic waste to completely decompose in nature is approximately 400 years, it can take much longer for other materials, such as glass, to decompose. For example, the time it takes for a glass bottle to decompose in nature is calculated to be approximately one million years [6, 7]. Therefore, the damage that recyclable materials (plastic, glass, metal, etc.) cause to nature has very serious and long-term consequences [8].

With the diversification of consumption habits and the rise in living standards, an increase in the amount of domestic waste is also observed. This increase has caused many large cities to face a waste pressure called “garbage siege”. Classification of waste is of great importance in terms of efficient use of resources and environmental sustainability [9–11]. However, this classification process cannot reach the desired level of efficiency due to low public awareness, diversity of

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waste types, and differences in local practices. Especially traditional separation systems, which are done manually, have significant disadvantages such as low efficiency, high labor cost, and health risks [3, 12, 13].

In order to overcome these problems and make recycling processes more efficient, many countries have made legal regulations and turned to technological solutions. However, waste separation is still largely based on human intervention [14, 15]. In recent years, the development of artificial intelligence, machine learning, and especially deep learning techniques has shown promise in terms of automating and optimizing this process [16–18]. Therefore, the integration of artificial intelligence-based technologies into waste management systems can provide significant benefits in terms of time, cost, and environmental health.

Deep learning, thanks to its achievements in the field of computer vision, is widely used in tasks such as image classification, object detection, and semantic segmentation [19–22]. In particular, Convolutional Neural Networks (CNN) have significantly increased accuracy rates by automating feature extraction processes [23–25]. These advances play a critical role in the process of automatic classification of waste more accurately and efficiently [26, 27].

Recent studies have shown that not only deep learning models but also supporting these models with various hyperparameter optimization techniques significantly increase the classification performance. For example, in the study [28], a deep learning architecture optimized with the Salp Swarm Algorithm (SSA) was presented, and thanks to this, the training time was shortened and the accuracy rates were improved. Similarly, in the study [29], the residual blocks developed with the Manta Ray Foraging Optimization (MRFO) algorithm were combined, the deep feature learning capacity of the model was increased, and high success was achieved in multi-class discrimination tasks. In addition, in the study [30], an optimized deep learning approach that works effectively on different types of images was developed, and high accuracy rates were achieved in multi-faceted classification tasks. Such optimized deep learning architectures are not limited to certain areas, but can also offer significant potential in environmental image analyses such as waste classification.

In this study, a multi-scale model designed using deep learning architectures, especially for the classification of environmental waste, is proposed. The features obtained from EfficientNetB0 and InceptionV3 architectures were combined with the HyperColumn approach, allowing for more effective use of both low-level and high-level visual representations. The aim of classifying these enriched features using XGBClassifier is to extract more detailed information from the waste images and increase the classification accuracy. In this context, our main contributions in this work are as follows:

- By combining EfficientNetB0, InceptionV3, and HyperColumn techniques, a collective learning approach was developed, which enabled effective learning of features at multiple resolutions.
- By combining the feature maps obtained from EfficientNetB0 and InceptionV3 using the HyperColumn technique, the proposed model was able to effectively process both general and detailed features by blending information from different layers.
- By using the XGBClassifier in the classification phase, the final classification performance was optimized, and the success rate was increased.
- It was revealed that the proposed model can be used effectively in areas such as environmental and waste management and that it can contribute to decision support systems in these areas.

2 Related works

There are many studies in the literature using deep learning and machine learning models in the classification of waste types. In the study conducted by Bircanoğlu et al. [31], the transfer learning method was used using CNN architectures such as Inception, ResNet, and DenseNet121. The TrashNet dataset was used in the study and a 95% success rate was achieved by considering data augmentation techniques. Chu et al. [32] proposed a hybrid deep learning model based on Multi-Layer Perceptron (MLP), AlexNet, and data augmentation techniques. A new dataset divided into two classes (recyclable and others) was created with camera images and a 98.2% success rate was achieved in their study. Liu et al. [26] achieved 98.9% accuracy on the Real Scene dataset with their proposed Depth-wise Separable Convolution Attention Module (DSCAM) method. Li et al. [33] proposed a multimodel cascaded convolutional neural network (MCCNN). MCCNN performs garbage detection by combining three subnetworks: DSSD, YOLOv4, and Faster-RCNN. The authors demonstrated that MCCNN performs the best by improving the detection accuracy by 10% on average.

Shi et al. [34] found that deep learning performed poorly in garbage image classification due to a lack of data and aimed to solve this problem by optimizing the network structure on the TrashNet dataset. The study showed that traditional deep networks and short-circuit connections were not effective enough on the TrashNet dataset. Instead, a method was developed that provides feature fusion by expanding the network branches and adding additional layers. The proposed M-b Xception network showed superiority over existing methods with an accuracy rate of 94.34% on the TrashNet dataset. In the study conducted by Adedeji and Wang [35], a system was developed in which a 50-layer residual network (ResNet-50) model was used as a CNN and this model served as a feature

extractor. The SVM technique was used to classify waste into different groups such as glass, metal, paper, and plastic. The proposed system was tested on the garbage image dataset developed by Gary Thung and Mindy Yang and achieved an accuracy rate of 87%.

Nowakowski and Pamua [36] aimed to develop an image recognition system for the automatic recognition and classification of e-waste from photographs. The system classified e-waste with accuracy ranging from 90 to 97% using deep learning-based CNN and R-CNN. Sukel et al. [37] achieved a maximum accuracy of 80% using the GIGO dataset. Mao et al. [38] aimed to develop an automatic classification robot that provides efficiency in recycling processes using the DenseNet121 model. The performance of DenseNet121 was optimized using data augmentation and genetic algorithm (GA). As a result, the optimized model exhibited a high performance with an accuracy rate of 99.6%. The contribution of data augmentation to the accuracy increase was more effective than optimizing the fully connected layer. It was observed that the model optimized with GA significantly improved the performance of DenseNet121.

In another study conducted by Yang and Thung [39] on the identification of waste, a computer vision method was developed to classify waste into six recycling categories: glass, paper, metal, plastic, cardboard, and garbage. For this purpose, a dataset containing 400–500 images was created for each class. The models used include SVM and CNN based on SIFT features. According to the experimental results, the SVM model performed better than CNN; however, CNN could not reach its full potential due to difficulties in optimizing hyperparameters. The authors achieved 63% accuracy with the dataset they named TrashNet. Panwar et al. [40], on the other hand, highlighted the dangers of water pollution and proposed a dataset based on the TACO dataset called AquaTrash. A deep learning-based object detection model called AquaVision was applied to the AquaTrash dataset to detect and classify pollutants and harmful waste floating in water bodies. The proposed model detects water pollution with an average precision of 81.48%. Another study by Vo et al. [41] proposed a model using deep neural networks to automatically classify garbage that can be implemented in smart waste sorting machines. The researchers generated the VN-trash dataset consisting of 5904 images of three different waste classes (Organic, Inorganic, Medical) from Vietnam. The study aims to improve the garbage classification performance by developing the DNN-TC model, which is an improvement of the ResNext model. Experiments show that DNN-TC achieves 94% and 98% accuracy rates on the TrashNet and VN-VN-trash datasets, respectively, thus outperforming existing methods.

3 Dataset

For the classification of waste images, Household_Garbage [42] and TrashNet [39] datasets were used. TrashNet [39] dataset was created by Yang and Thung to classify waste into six recycling categories: glass, paper, metal, plastic, cardboard, and garbage. This dataset contains 2527 garbage images in total, including 403 cardboard, 501 glass, 410 metal, 594 paper, 482 plastic, and 137 trash images.

The Household_Garbage dataset created by Mohamed [42] consists of 12 classes including paper, cardboard, biological, metal, plastic, green-glass, brown-glass, white-glass, clothes, shoes, battery, and trash images. This dataset contains 15,515 images in total, including 945 batteries, 985 biological, 607 brown-glass, 891 cardboard, 5325 clothes, 629 green-glass, 769 metal, 1050 paper, 865 plastic, 1977 shoes, 697 trash and 775 white-glass. Some sample images from the TrashNet and Household_Garbage datasets are presented in Fig. 1.

4 Method

In this study, to classify the waste images, the intermediate features obtained from the EfficientNetB0 and InceptionV3 architectures were combined using the HyperColumn technique, resulting in multi-resolution, rich feature representations. In the preprocessing stage, the waste images were resized to $224 \times 224 \times 3$ and fed as input to both architectures. After the feature maps extracted from specific layers of both models were combined under the HyperColumn structure, the GlobalAveragePooling operation was applied to support general feature learning by reducing the spatial dimensions. The obtained features were then passed through four different Fully Connected Layer (FCL) stages in the Feature Processing Layers. Finally, the features from the FCL were classified using the XGBClassifier algorithm. The general block diagram of the proposed model is shown in Fig. 2.

4.1 InceptionV3

InceptionV3 [43] is the third version of the Inception architecture, a popular image classification model in the field of deep learning. This model is designed to be used in computer vision applications, especially by providing high accuracy on large image datasets. The model is designed using symmetric and asymmetric building blocks, and each block consists of various components. These components include convolutional layers, average and max pooling layers, concatenation operations, dropout layers, and fully connected layers. This multi-layer structure optimizes the overall performance by allowing the model to learn more complex and high-dimensional features.

Fig. 1 Some sample images found in the datasets

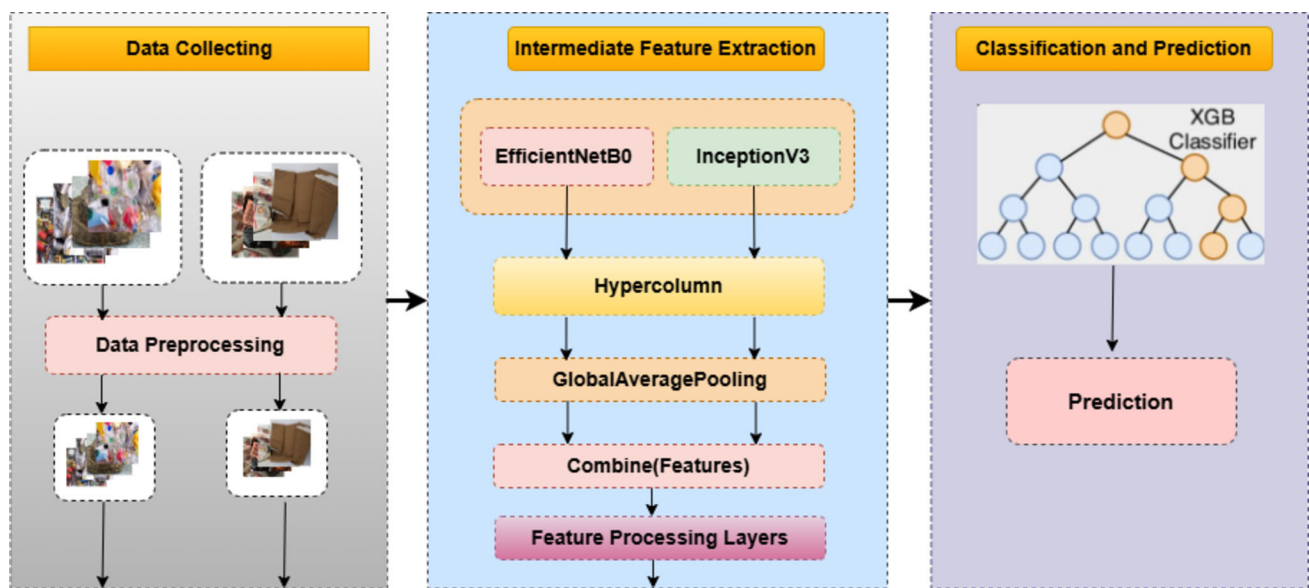
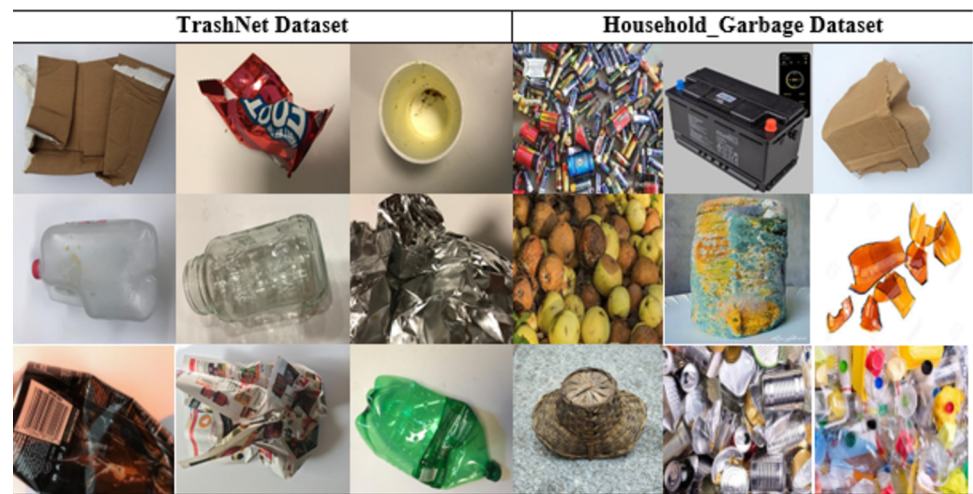
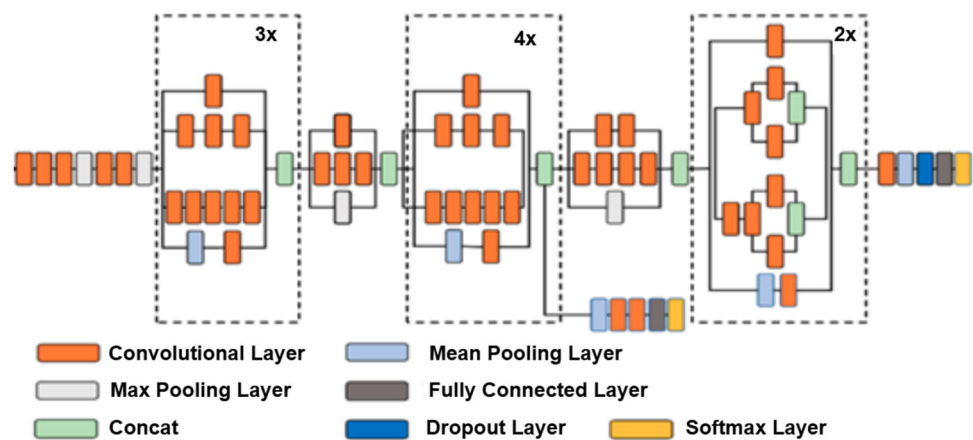


Fig. 2 Overall block diagram of the proposed model

Fig. 3 The basic structure of the Inceptionv3 model



InceptionV3 uses wider architectures with more layers than previous versions. These layers help the model learn deeper features. The Inception architecture allows the model to capture both low and high-resolution features by using convolutional filters of different sizes simultaneously. This contributes to learning more complex visual features. InceptionV3 reduces the computational load by splitting a 5×5 convolution into two separate 3×3 convolutions with factorized convolutions. To make the model learn better, small classifiers are used in the intermediate layers. These auxiliary classifiers are additional targets used during the training of the main model [43–46]. The basic structure of the Inceptionv3 architecture is presented in Fig. 3.

4.2 EfficientNetB0

The biggest innovation of EfficientNet is its scaling strategy. While traditional methods usually scale the depth (number of layers), width (number of filters in layers), or image resolution of the model separately, EfficientNet uses a technique called compound scaling. This technique scales both the depth and width of the model and the input image resolution in a balanced way. EfficientNetB0 is the base level of this strategy and is smaller and faster compared to larger EfficientNet models. EfficientNetB0 [47] uses a new activation function called Swish, instead of the standard ReLu (Rectified Linear Unit) activation function. Swish is a nonlinear activation function and helps the model achieve better results by increasing its learning capacity. This model uses layers called MBConv, which improve both efficiency and performance. MBConv layers combine compression and expansion operations. This increases the information density of the model, while reducing unnecessary computations, allowing it to run faster and more efficiently [48, 49]. The basic structure of the EfficientNetB0 architecture is given in Fig. 4.

4.3 HyperColumn

The term HyperColumn is a concept that has its origins in neuroscience and has been adapted to CNN based on its functional role in the brain structure. In the context of CNN, the HyperColumn technique is defined as the process of combining the features of each pixel in all layers of the model when an image is given as input to the model [50]. For each pixel, a HyperColumn vector is created that includes all the features extracted at each stage of the model. With this method, the most important feature in the created vectors is selected and transferred to the next layers. This technique is based on activation maps; hypervectors are created by combining the features obtained from these maps on a pixel basis [51, 52]. In the final stage, the HyperColumn technique selects the most effective feature from the vectors created for each pixel and transmits it to the deep layers of the network. The feature

vector at the i th position in the HyperColumn technique is shown below.

$$f_k = \sum_k \alpha_{ik} F_k \quad (1)$$

Here, F denotes the feature map and f denotes the upsampled feature map. α_{ik} denotes the location of the parameters i and k [51, 52].

In the proposed model, the HyperColumn technique aims to increase the performance of the model by combining the pixel-level features extracted from the Inceptionv3 and EfficientNetB0 methods.

4.4 XGBClassifier

XGBClassifier [53] is an ensemble learning model used in classification problems. Ensemble learning methods allow multiple weak models (e.g., decision trees) to be combined to make a general prediction. XGBClassifier, a special optimization of the gradient boosting (GB) algorithm [54], performs classification using decision trees. Each new tree is trained to correct the errors of previous trees. This process is optimized with gradient descent and the prediction errors are minimized [53, 55]. The stages of the GB algorithm on which the XGBClassifier algorithm is based are as follows.

- Determine the initial model: First, a simple estimate is made for the training data set. This can usually be the mean of the data set.
- Calculate the residual values: In each new iteration, the errors made by the model (residuals) are calculated as follows.

$$r_i = y_i - \hat{y}_i \quad (2)$$

Here, r_i is the residual, y_i is the actual value and $\hat{y}_i$ is the value predicted by the model.

- Creating a new model: In each new iteration, a new model is added to correct the errors. This model is trained on the residuals. XGBClassifier uses gradient descent to minimize the errors:

$$\hat{F}_{m+1}(x) = \hat{F}_m(x) - \gamma h_m(x) \quad (3)$$

Here $h_m(x)$ represents the m th model and γ represents the learning rate. The new model is optimized to minimize the errors of the previous model.

- Prediction: After all trees are trained, the final prediction of the model is the weighted sum of the results of each tree.

4.5 Proposed model

The proposed model for the classification of waste images is designed based on the ensemble learning approach that

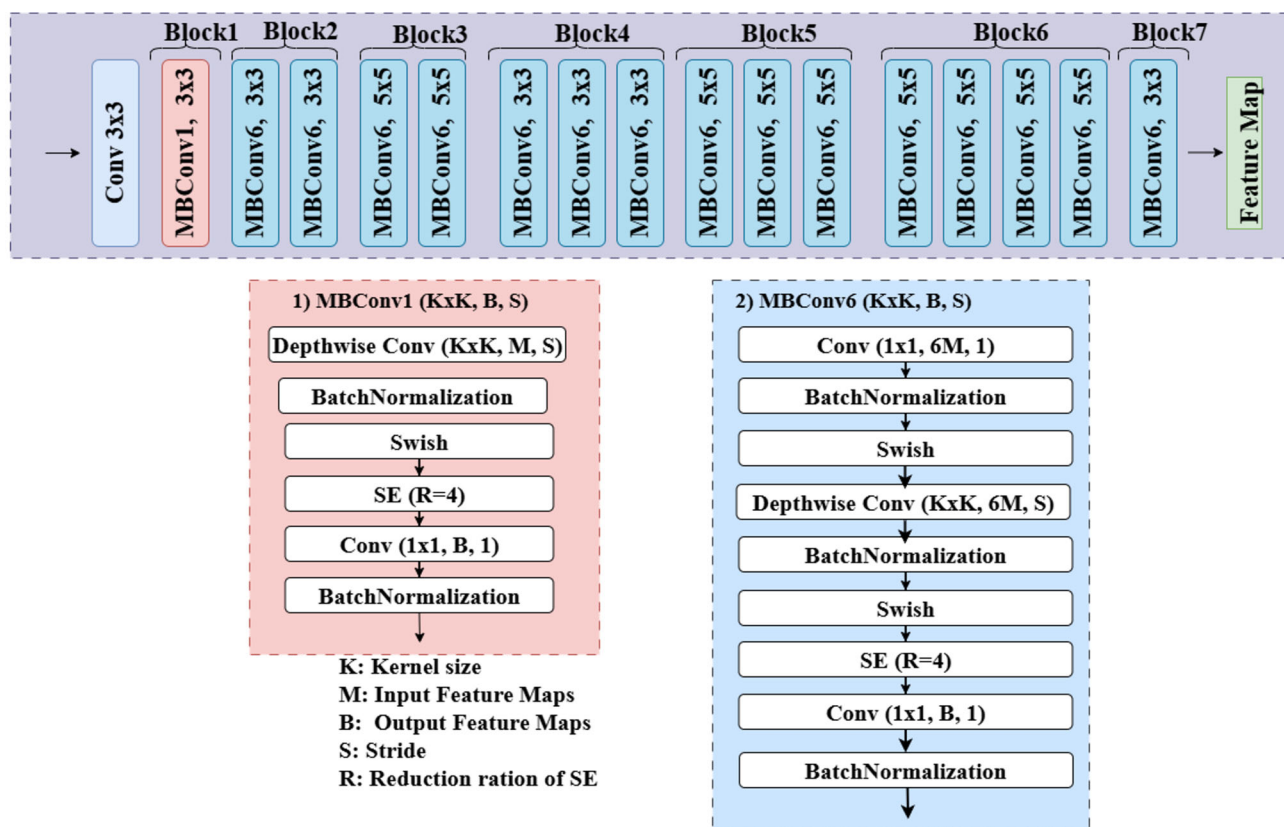


Fig. 4 The basic structure of the EfficientNetB0 model

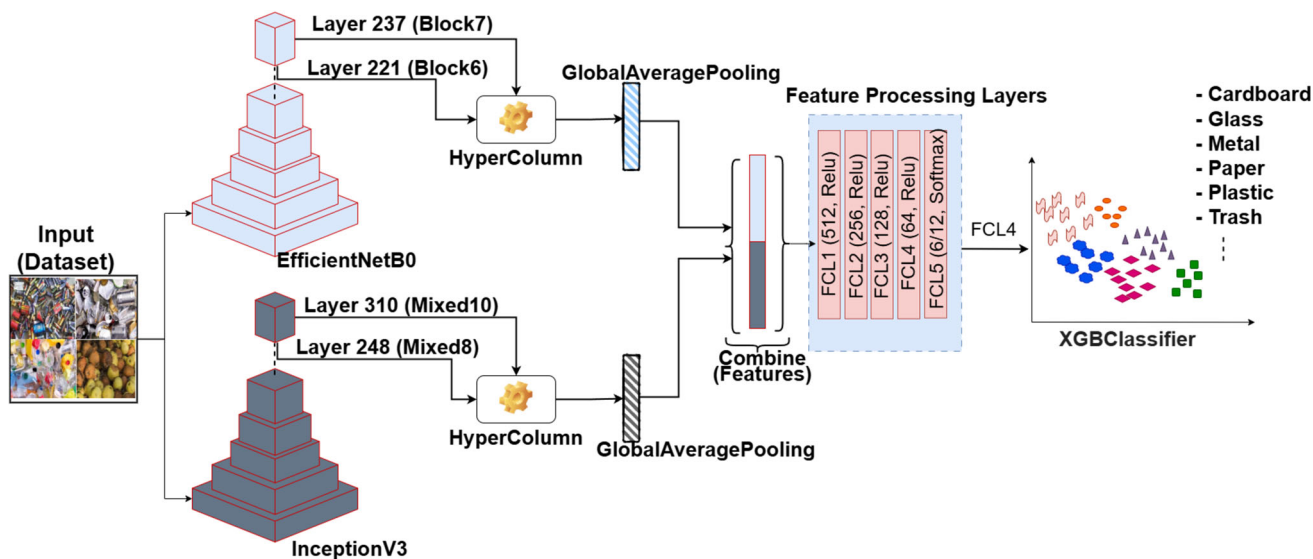


Fig. 5 The basic structure of the proposed model

includes EfficientNetB0, InceptionV3, and HyperColumn techniques. The basic structure of the proposed model is given in Fig. 5. The main purpose of this model is to classify waste images with different resolutions more effectively. EfficientNetB0 and InceptionV3 architectures are known as architectures that provide high accuracy and efficiency among deep learning models. While EfficientNetB0 achieves high performance with fewer parameters by scaling the model size, InceptionV3 captures multi-scale features of images using filters of different sizes. Simultaneous training of these architectures allows the model to learn various features better.

First of all, waste images are converted to $224 \times 224 \times 3$ size and given to the input of both architectures. In the EfficientNetB0 model, the feature maps obtained from the last convolution layers of layers 221 to 237 in Block6 and Block7 are combined with the HyperColumn technique. The HyperColumn technique provides multi-resolution information by combining features from different layers, thus enabling the model to learn richer features. Similarly, in the InceptionV3 architecture, feature maps obtained from layers 248 and 310 are combined with the HyperColumn technique. The features combined with the HyperColumn technique were subjected to the GlobalAveragePooling process to enable the model to learn more effective and detailed features. The GlobalAveragePooling reduces overfitting by reducing spatial dimensions and encourages the model to learn more general features. These obtained features are passed through five Fully Connected Layers (FCL) with 512, 256, 128, 64, and 6/12 neurons. The FCL layers allow the model to process the learned features at a higher level and prepare them for the classification process. Relu was used as the activation function after each FCL process. ReLu increases the learning capacity by adding nonlinear features to the model and enables the model to learn more complex relationships [56, 57].

Finally, the classification process is performed with the XGBClassifier technique. XGBClassifier improves the overall performance of the model as a method that can provide higher accuracy compared to traditional classification techniques. This ensemble learning approach aims to provide higher classification accuracy and efficiency by using different architectures and techniques together.

The proposed method is based on combining the mid-level representations obtained from EfficientNetB0 and InceptionV3 architectures and giving this combined representation to the XGBClassifier classifier. The mathematical formulation and process of the proposed model can be described step by step as follows:

Step 1: Extracting intermediate representation features:

For a given input image $x \in \mathbb{R}^{H \times W \times C}$, features are extracted from the specified intermediate layers of EfficientNetB0 and InceptionV3 models:

$$\begin{aligned} fEffB0(x) &\in \mathbb{R}^{d_1} \\ fIncV3(x) &\in \mathbb{R}^{d_2} \end{aligned} \quad (4)$$

Here, d_1 and d_2 represent the dimensions of feature vectors extracted from EfficientNetB0 and InceptionV3, respectively.

Step 2: Combining features (Hypercolumn).

The representations obtained from both models are combined using the Hypercolumn technique. This process allows the combination of features from different layers of both networks:

$$h(x) = Hypercolumn(fEffB0(x), fIncV3(x)) \in \mathbb{R}^{d_1+d_2} \quad (5)$$

Here, Hypercolumn refers to the combination of features extracted from different layers of both models.

Step 3: Classification Process.

The combined feature vector is given as input to the XGBClassifier classifier to obtain the predicted class label $\hat{y}$:

$$\hat{y} = XGB(h(x)) \quad (6)$$

Here, $XGB(.)$ represents a gradient boosting tree-based classifier function.

As a result, the general mathematical formula of the proposed model can be expressed as follows:

$$\hat{y} = XGB(Hypercolumn(fEffB0(x), fIncV3(x))) \in \mathbb{R}^{d_1+d_2} \quad (7)$$

5 Performance evaluation metrics

In this study, accuracy (Acc), precision (Pre), recall (Re), f1-Score (F1), specificity (Spe), AUC (Area Under the Curve) metrics, and confusion matrix were used for the performance analysis of the models [58–60].

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} * 100 \quad (8)$$

$$Pre = \frac{TP}{TP + FP} * 100 \quad (9)$$

$$Re = \frac{TP}{TP + FN} * 100 \quad (10)$$

$$F1 = \frac{2 * Pre * Rec}{Pre + Rec} * 100 \quad (11)$$

$$Spe = \frac{TN}{TN + FP} * 100 \quad (12)$$

$$TPR = \frac{TP}{TP + FN} \quad (13)$$

$$FPR = \frac{FP}{FP + TN} \quad (14)$$

$$AUC = \int_0^1 TPRd(FPR) * 100 \quad (15)$$

Here TP (True-Positive), TN (True-Negative), FP (False-Positive), and FN (False-Negative) show the values used in the confusion matrix. Also, TPR represents the true positive rate and FPR represents the false positive rate.

6 Experimental studies and their results

In this section, all experimental stages related to the training, testing, and comparative evaluation of the proposed model are explained in detail. TrashNet and Household_Garbage datasets were used for the classification of garbage images. Two different experimental studies were conducted on each dataset. In the first experimental application, a comparative analysis was made with the EfficientNetB0 and InceptionV3 architectures that form the basis of the proposed model, and the performance improvement of the model was revealed. In addition, as a result of this experimental study, the class-wise accuracy of the proposed model was analyzed. In the second experimental application, the success of the proposed model with current CNN-based architectures and Transformer-based approaches was analyzed.

6.1 Environment, tools and dataset

The experimental studies in this study were carried out in Google Colaboratory using the Python programming language. TensorFlow and Keras deep learning libraries were used in the model development processes. Adam optimization algorithm and categorical cross-entropy loss function were used for training the models. In addition, the batch size value was determined as 128, and the models were trained for 40 epochs.

80% of the TrashNet and Household_Garbage datasets used for the classification of waste images were separated for training, and 20% for testing. Details regarding the class distribution in the training and test datasets are presented in Table 1 and Table 2, respectively.

The waste images in the TrashNet dataset were labeled as cardboard $\rightarrow$ 0, glass $\rightarrow$ 1, metal $\rightarrow$ 2, paper $\rightarrow$ 3, plastic

$\rightarrow$ 4, and trash $\rightarrow$ 5. Similarly, the classes in the Household_Garbage dataset are labeled as follows: battery $\rightarrow$ 0, biological $\rightarrow$ 1, brown-glass $\rightarrow$ 2, cardboard $\rightarrow$ 3, clothes $\rightarrow$ 4, green-glass $\rightarrow$ 5, metal $\rightarrow$ 6, paper $\rightarrow$ 7, plastic $\rightarrow$ 8, shoes $\rightarrow$ 9, trash $\rightarrow$ 10 and white-glass $\rightarrow$ 11. These class labels were used as the basis for the model training and evaluation processes.

6.2 Results and analysis for the trashNet dataset

The general performance results of the first experimental study conducted with the TrashNet dataset are presented in Table 3, and the confusion matrix showing class-based accuracies is presented in Fig. 6. In addition, detailed success metrics of the proposed model at the class level are given in Table 4.

According to the findings in Table 3, the proposed model has demonstrated higher success in all metrics (Accuracy, Precision, Recall, F1-score, Specificity, and AUC) compared to the EfficientNetB0 and InceptionV3 models. It is seen that the performance of the architectures with the proposed model has increased between 1.9% and 13.24%.

We can say that this success is due to the combination of the parameter-efficient structure of EfficientNetB0, which forms the basic structure of the proposed model, and the multi-scale feature extraction capacity of InceptionV3 with the HyperColumn technique. With the HyperColumn approach, the model was able to evaluate features from different depths together, and it was possible to learn both low- and high-level information effectively. Thus, the model produced richer representations, strengthened the distinction between classes, and significantly increased the overall accuracy rate.

When the confusion matrix (Fig. 6) results of the architectures given for the first experimental study are examined, it is seen that the proposed model does not make any incorrect classifications in the Cardboard (0), Metal (2), and Trash (5) classes. Only 1 Plastic (4) sample was incorrectly classified in the Glass (1) class, and 1 Cardboard (0) sample made an error in the Paper (3) class. 1 Glass (1) sample was also incorrectly assigned in the Plastic (4) class. The proposed model correctly classified most classes, but we can say that it made small errors among similar classes. The high prediction rate of the proposed model shows that the model can discriminate well between classes and has high generalization ability on the test data.

The class success results of the proposed model on the TrashNet dataset are given in Table 4. According to the results presented in Table 4, the class success rates of the proposed model on the TrashNet dataset are quite remarkable. Especially, the accuracy rate of 100% in the metal and trash classes shows that the model can distinguish these classes perfectly. The success rate of over 99% in other classes shows that

Table 1 Class distribution of training and test data for the TrashNet dataset

Dataset	Class	Train	Test	Total
TrashNet	Cardboard (0)	324	79	403
	Glass (1)	384	117	501
	Metal (2)	329	81	410
	Paper (3)	490	104	594
	Plastic (4)	379	103	482
	Trash (5)	115	22	137
Total		2021	506	2527

Table 2 Class distribution of training and test data for the Household_Garbage dataset

Dataset	Class	Train	Test	Total
Household_Garbage	Battery (0)	736	209	945
	Biological (1)	801	184	985
	Brown-glass (2)	491	116	607
	Cardboard (3)	693	198	891
	Clothes (4)	4292	1033	5325
	Green-glass (5)	500	129	629
	Metal (6)	624	145	769
	Paper (7)	822	228	1050
	Plastic (8)	686	179	865
	Shoes (9)	1585	392	1977
	Trash (10)	579	118	697
	White-glass (11)	603	172	775
Total		12,412	3103	15,515

Table 3 Test results of the proposed model and EfficientNetB0 and InceptionV3 models using the TrashNet dataset

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
EfficientNetB0	89.72	88.12	90.40	89.04	97.93	94.10
InceptionV3	87.55	85.55	87.24	86.27	97.48	92.36
Proposed Model	99.40	99.49	99.54	99.51	99.88	99.71

The bolded values indicate the best performance for each evaluation metric

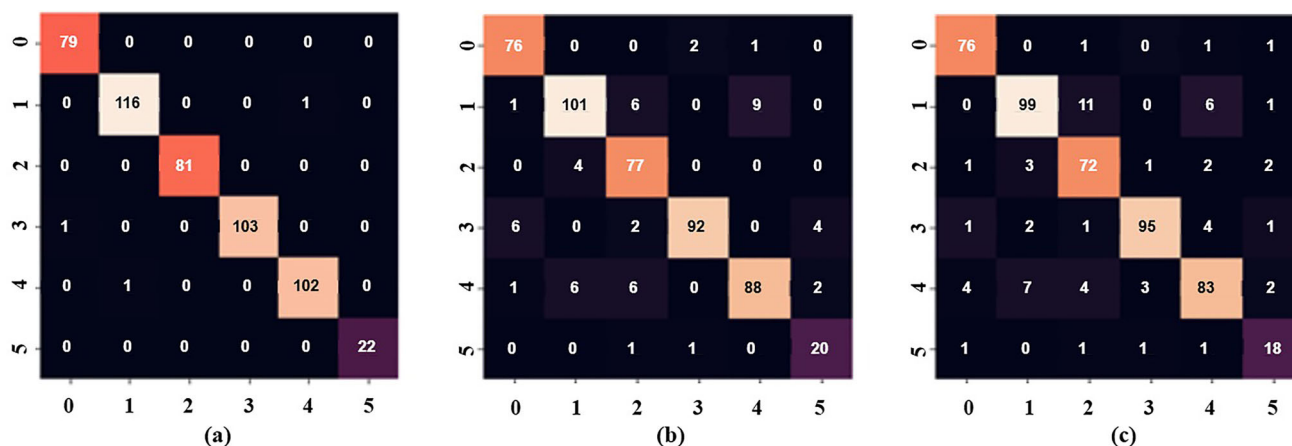
**Fig. 6** Confusion matrix results of **a** the proposed model, **b** EfficientNetB0, and **c** InceptionV3 models

Table 4 Class success results of the proposed model based on test images in the TrashNet dataset

Model	Waste Type	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
Proposed Model	Cardboard	100.00	98.75	100.00	99.37	99.77	99.88
	Glass	99.15	99.15	99.15	99.15	99.74	99.44
	Metal	100.00	100.00	100.00	100.00	100.00	100.00
	Paper	99.04	100.00	99.04	99.52	100.00	99.52
	Plastic	99.03	99.03	99.03	99.03	99.75	99.39
	Trash	100.00	100.00	100.00	100.00	100.00	100.00

the model has a very strong performance in general and can classify different types of waste correctly.

These results are an important indicator of the generalization ability of the model and how well it maintains the class balance. The accuracy of 100% in the metal and trash classes proves that the model has learned the features of these classes very well and can clearly distinguish between the classes. The accuracy of over 99% in other classes reveals the strong generalization capacity of the model and these accuracy rates show that the model does not experience overfitting problems.

The results of the experimental application performed to compare the performance of the proposed model with other models are presented in Table 5. When the obtained results are examined, it is seen that the proposed model shows a remarkable superiority in all metrics. The proposed model reached 99.40% accuracy, 99.49% precision, 99.54% recall, 99.51% F1 score, 99.88% specificity, and 99.71% AUC values. These values show that the success of the model in the classification task is quite high, and the correct classification rate is at a highly satisfactory level.

In particular, the high success of the proposed model in important metrics such as precision and recall reveals the ability of the model to correctly identify each class. This shows that the model is trained sensitively to the characteristics of different classes in the dataset and can distinguish each class with high accuracy. In addition, high values in additional metrics such as specificity and AUC emphasize that the overall performance of the model is reliable and strong.

On the other hand, considering the results of the comparative models, it is possible to say that the proposed model is significantly superior to other models not only in terms of accuracy but also in other classification metrics. This shows that the proposed model is more effective and efficient in solving the classification problem on the TrashNet dataset compared to other existing models.

6.3 Results and analysis for the household_garbage

The results of the first experimental study using the Household_Garbage dataset are presented in Table 6. The confusion

matrix results regarding the classification performance of the proposed model on the test data are given in Fig. 7, and the class success rates are given in Table 7. When the obtained results are examined, it is seen that the proposed model exhibits a very high performance on this dataset and is similar to its performance on the TrashNet dataset. The proposed model, consisting of EfficientNetB0 and InceptionV3 architectures, produced superior results in all metrics compared to these two basic architectures. The proposed model provided a performance increase ranging from 0.49% to 16.97% according to different evaluation criteria.

These differences reveal that the model can extract stronger features, is more successful in data representation, and can distinguish between classes more effectively. In addition, it can be said that the combination strategy used in the architectural design of the model provides a more balanced classification by increasing the learning capacity and, at the same time, reducing the risk of overfitting.

Figure 7 presents the class-wise accuracy results of the architectures using the Household_Garbage test dataset. Upon examining the results, it is observed that the proposed model demonstrates exceptionally high accuracy performance across all classes. The model accurately predicted the images in the battery (0), brown-glass (2), cardboard (3), clothes (4), green-glass (5), metal (6), paper (7), shoes (9), and trash (10) classes without any errors. Similarly, in the biological (1) class, only one sample was misclassified as green-glass (5). Additionally, in the white-glass (11) class, only two samples were predicted as metal (6). For the plastic (8) class, only one sample was misclassified. These results indicate that the proposed model achieves high accuracy in inter-class discrimination and provides strong generalization performance across the dataset.

The class success results of the proposed model in the Household_Garbage dataset according to the test images are presented in Table 7. It shows that the proposed model exhibits excellent classification performance in almost all classes except for certain classes in the Household_Garbage dataset. Especially, 100% success in all classes except for biological, green-glass, metal, plastic, and white-glass classes

Table 5 Test results of the models used according to the TrashNet dataset

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
MobileNetV2	70.36	61.07	62.01	60.32	93.96	83.58
DenseNet121	88.53	88.51	88.14	88.29	97.64	92.89
Xception	88.93	88.13	88.30	97.84	97.74	93.05
EfficientNetB0	89.72	88.12	90.40	89.04	97.93	94.10
InceptionV3	87.55	85.55	87.24	86.27	97.48	92.36
ViT	94.46	94.67	91.20	92.54	98.85	95.03
SwinTransfrom	90.90	91.41	88.98	89.79	98.14	93.56
Proposed Model	99.40	99.49	99.54	99.51	99.88	99.71

The bolded values indicate the best performance for each evaluation metric

Table 6 Test results of the proposed model using the Household_Garbage dataset and the EfficientNetB0 and InceptionV3 models

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
EfficientNetB0	94.61	93.14	92.34	92.67	99.50	95.92
InceptionV3	88.14	87.46	82.84	84.44	98.88	90.86
Proposed Model	99.87	99.77	99.81	99.79	99.99	99.90

The bolded values indicate the best performance for each evaluation metric

proves that the model can successfully distinguish a large part of them.

However, the failure to achieve 100% success in some classes (especially biological, green-glass, metal, plastic, and white-glass) may indicate that these classes are more difficult for the model to distinguish than others or that the images belonging to these classes contain more variety and complexity. Among these classes, materials such as biological waste and glass may have more similarities in terms of visual features, which may cause the model to be confused.

In the second experimental study conducted using the Household_Garbage dataset, the performance of different architectures was analyzed. The corresponding test results are presented in Table 8. Upon examining the results, it is observed that the proposed model achieved the highest performance with 99.87% accuracy, 99.77% precision, 99.81% recall, 99.79% F1-Score, 99.99% Specificity, and 99.90% AUC. These results demonstrate that the proposed model exhibits exceptionally high performance.

Especially the 99.87% accuracy rate shows that the model works almost without error in the classification task on the test data. The 99.77% precision value indicates how much of the data the model predicts as positive is correct, while the 99.81% recall rate shows how well the model captures positive classes. The closeness of these two criteria shows that the model exhibits a balanced performance. The F1-score of 99.79% highlights the high harmonic mean of precision and recall, while 99.99% Spe reveals that the model has an excellent capacity to correctly recognize negative classes. Finally, the 99.90% AUC is another important success indicator that

summarizes the overall performance of the model's classification ability. AUC close to 1 proves that the model is quite successful in the classification task.

The prediction results of some sample images of the proposed model using the Household_Garbage dataset are given in Fig. 8. When the images are examined, it is seen that the predictions are quite successful. The real classes and the predicted classes are given on the image.

6.4 General accuracy results of architectures

The overall accuracy results of the architectures are given in Fig. 9. The proposed model achieved the highest accuracy rates on both datasets (TrashNet and Household_Garbage) and outperformed all other architectures. In particular, it outperformed both CNN-based models (e.g., MobileNetV2, DenseNet121, EfficientNetB0) and Transformer-based architectures (e.g., ViT, Swin Transformer) with 99.4% accuracy rates on the TrashNet dataset and 99.87% accuracy rates on the Household_Garbage dataset.

While the Transformer-based ViT model gave more successful results than other classical architectures with 94.46% on TrashNet and 95.93% on Household_Garbage, the proposed model exceeded this success with differences exceeding approximately 5%. Lighter architectures, such as MobileNetV2, which have especially low accuracy rates, were insufficient in complex data structures.

These results reveal that the proposed model has a stronger representation learning capacity and generalization ability in distinguishing both simple and complex waste types. Thus,

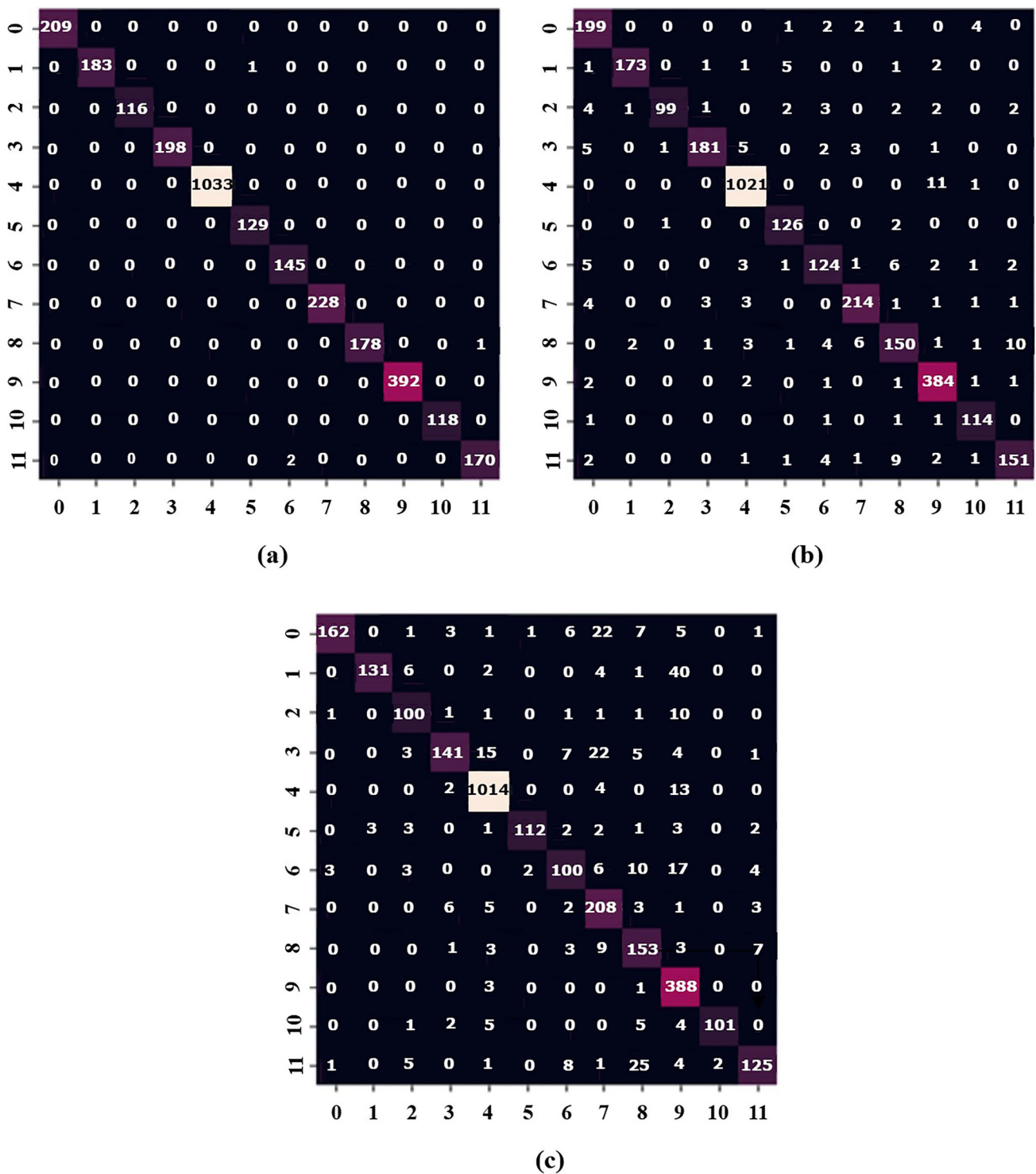


Fig. 7 Confusion matrix results of **a** the proposed model, **b** EfficientNetB0, and **c** InceptionV3 models based on test images in the Household_Garbage dataset

Table 7 Class success results of the proposed model in the Household_Garbage dataset according to the test images

Model	Waste Type	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
Proposed Model	Battery	100.00	100.00	100.00	100.00	100.00	100.00
	Biological	99.46	100.00	99.46	99.73	100.00	99.73
	Brown-glass	100.00	100.00	100.00	100.00	100.00	100.00
	Cardboard	100.00	100.00	100.00	100.00	100.00	100.00
	Clothes	100.00	100.00	100.00	100.00	100.00	100.00
	Green-glass	100.00	99.23	100.00	99.61	99.97	99.98
	Metal	100.00	98.64	100.00	99.32	99.93	99.97
	Paper	100.00	100.00	100.00	100.00	100.00	100.00
	Plastic	99.44	100.00	99.44	99.72	100.00	99.72
	Shoes	100.00	100.00	100.00	100.00	100.00	100.00
	Trash	100.00	100.00	100.00	100.00	100.00	100.00
	White-glass	98.84	99.42	98.84	99.13	99.97	99.40

Table 8 Test results of the models based on the Household_Garbage dataset

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
MobileNetV2	78.47	77.94	73.93	73.60	98.05	85.99
DenseNet121	91.27	88.29	88.09	87.77	99.22	93.66
Xception	92.23	89.59	90.02	89.29	99.30	94.66
EfficientNetB0	94.61	93.14	92.34	92.67	99.50	95.92
InceptionV3	88.14	87.46	82.84	84.44	98.88	90.86
ViT	95.93	94.38	95.77	94.91	99.64	97.71
SwinTransfrom	91.45	88.96	90.68	89.31	99.23	94.96
Proposed Model	99.87	99.77	99.81	99.79	99.99	99.90

The bolded values indicate the best performance for each evaluation metric

thanks to both the depth and multi-feature fusion approach, it is seen that the proposed model provides a significant advantage in overall accuracy.

6.5 Complexity analysis of architectures

The parameter counts and time complexities during the training and testing phases of different deep learning models were comparatively analyzed. As shown in Table 9, the MobileNetV2 and EfficientNetB0 models stand out with their low parameter counts (2 M and 4 M, respectively), and consequently, short training (2.17 and 3.54 min) and testing times. On the other hand, the ViT model has by far the highest model size with 397 M parameters, which results in significantly longer training time (83.43 min) and especially CPU-based testing time (35.42 min). Another transformer-based model, Swin Transformer, also has a large model size with 87 M parameters and exhibits relatively higher computational costs during both training and testing phases.

The proposed model has a moderate size with 28 M parameters. Although its training time (7.29 min) is slightly longer

compared to some classical CNN architectures, it is considerably shorter than large-scale models such as ViT and Swin Transformer. The GPU testing time (0.26 min) is fast, and the CPU testing time (1.25 min) is also at a balanced level. These results demonstrate that the proposed model establishes an effective trade-off between computational efficiency and model capacity.

In conclusion, while low-parameter models (e.g., MobileNetV2 and EfficientNetB0) offer advantages in resource-constrained environments that require fast inference, the proposed model provides an optimized solution between accuracy and efficiency with reasonable computational cost and higher parameter capacity.

7 Ablation study

In this section, an ablation study was conducted to examine the effects of feature extraction at different levels on model performance. To this end, the contribution of intermediate representation features extracted from various layers of the

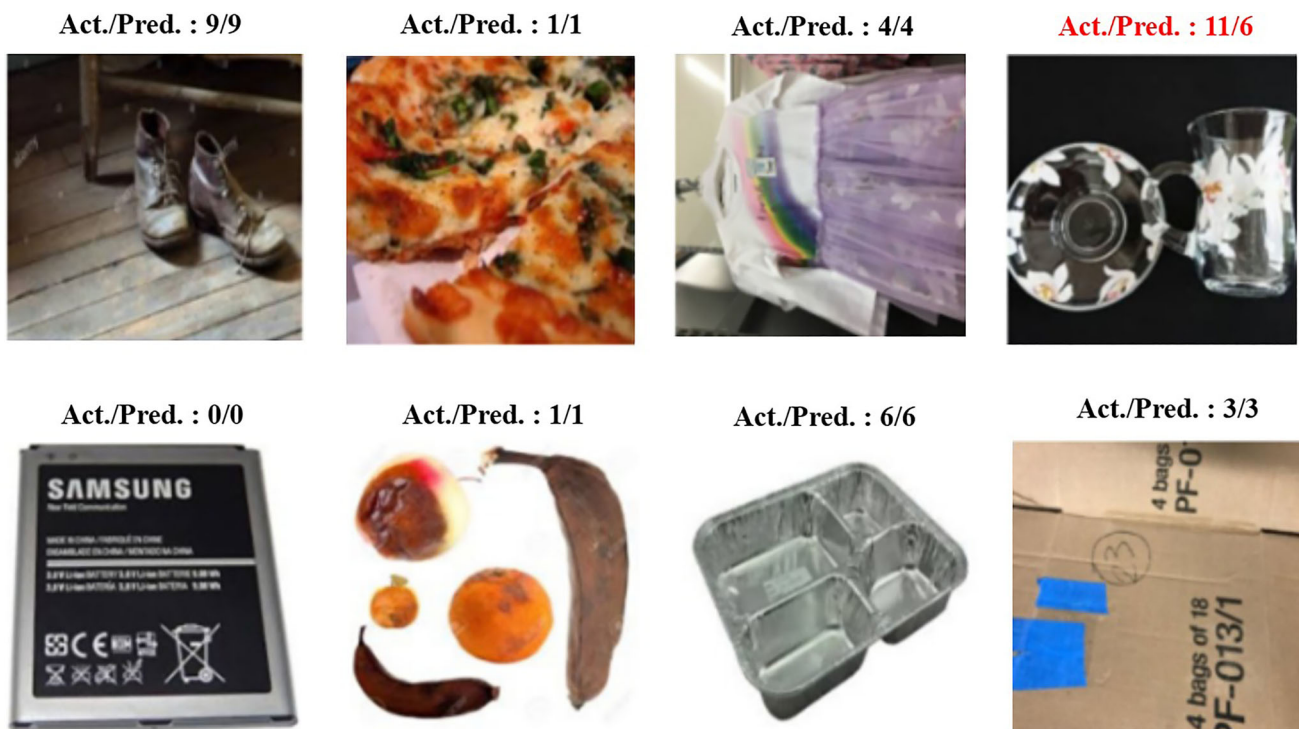


Fig. 8 Prediction of some sample images with the proposed model

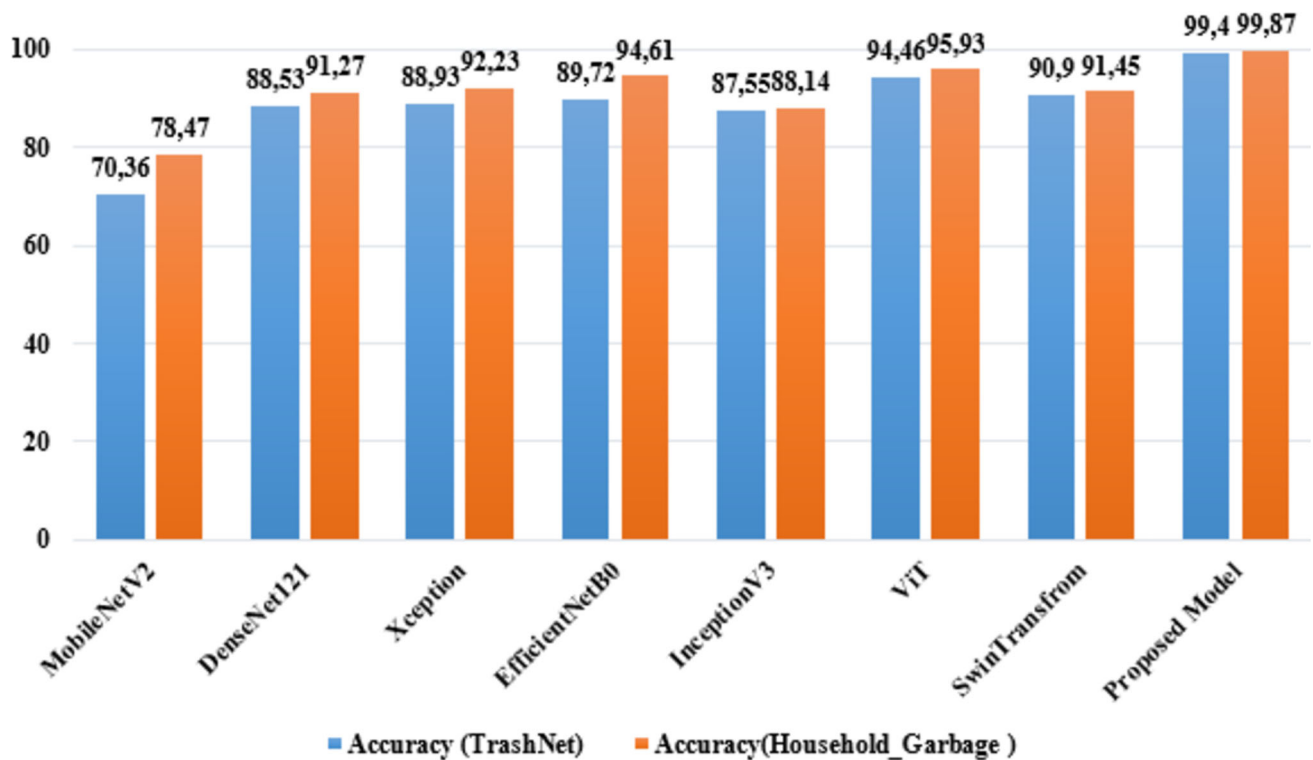


Fig. 9 Accuracy rates of the models based on test images in TrashNet and Household_Garbage datasets

Table 9 Comparative analysis of the number of parameters and computational complexity of the models

Method	Number of parameters	Train time with GPU (min.sec.)	Test time with GPU (min.sec.)	Test time with CPU (min.sec.)
MobileNetV2	2M	2.17	0.7	0.27
DenseNet121	7 M	4.33	0.15	1.49
Xception	20 M	6.20	0.21	1.48
EfficientNetB0	4 M	3.54	0.13	0.43
InceptionV3	21 M	6.48	0.18	0.59
ViT	397 M	83.43	2.25	35.42
SwinTransfrom	87 M	21.28	1.41	15.13
Proposed Model	28M	7.29	0.26	1.25

Table 10 Features of the developed models

Model Name	EfficientNetB0 layers	InceptionV3 layers
Model_1	block4c_add (Add)(119) block5c_add (162)	mixed6 (196) mixed7 (228)
Model_2	block3a_bn (52), block3b_add (Add) (75)	mixed4 (132) mixed5 (164)
Model_3	block2a_bn (23) block2b_add (46)	mixed1 (63) mixed2 (86)
Model_4	stem_bn (6) block2a_expand_activation (20)	conv2d_471 (4) activation_472(9)
Proposed Model	block6d_add (221) top_activation (237)	mixed8 (248) mixed10 (310)

EfficientNetB0 and InceptionV3 architectures to classification accuracy was analyzed individually. For each model, features obtained from different layer depths were used to determine at which stage more effective representations are learned.

Therefore, five different models were developed using intermediate layers extracted at various levels from EfficientNetB0 and InceptionV3. The main objective of these models is to leverage the representational power of specific layers in both architectures to enable more effective feature extraction. A summary of the developed models is provided in Table 10.

- **Model_1** uses features from the deeper blocks (block4c_add and block5c_add) of EfficientNetB0 and the mid-to-high-level mixed layers (mixed6 and mixed7) of InceptionV3. This approach aims to capture more abstract and conceptual representations.
- **Model_2** focuses on the mid-level blocks of EfficientNetB0 (block3a_bn, block3b_add) and the mid-level mixed layers of InceptionV3 (mixed4, mixed5). The goal of this model is to combine both low-level edge and texture information with mid-level conceptual features.

- **Model_3** extracts features from the early-stage layers of EfficientNetB0 (block2a_bn, block2b_add) and the early-mid mixed layers of InceptionV3 (mixed1, mixed2). This model is intended to capture more basic and detailed features.
- **Model_4** is built upon the initial layers of EfficientNetB0 (stem_bn, block2a_expand_activation) and the early convolution and activation layers of InceptionV3 (conv2d_471, activation_472). Thus, the model aims to maximize the capture of low-level edge, color, and basic texture information.
- The **Proposed Model** extracts features from the higher-level final blocks of EfficientNetB0 (block6d_add, top_activation) and the most advanced mixed layers of InceptionV3 (mixed8, mixed10). This architecture aims to enhance high-level representation by encompassing both complex patterns and conceptual classification information.

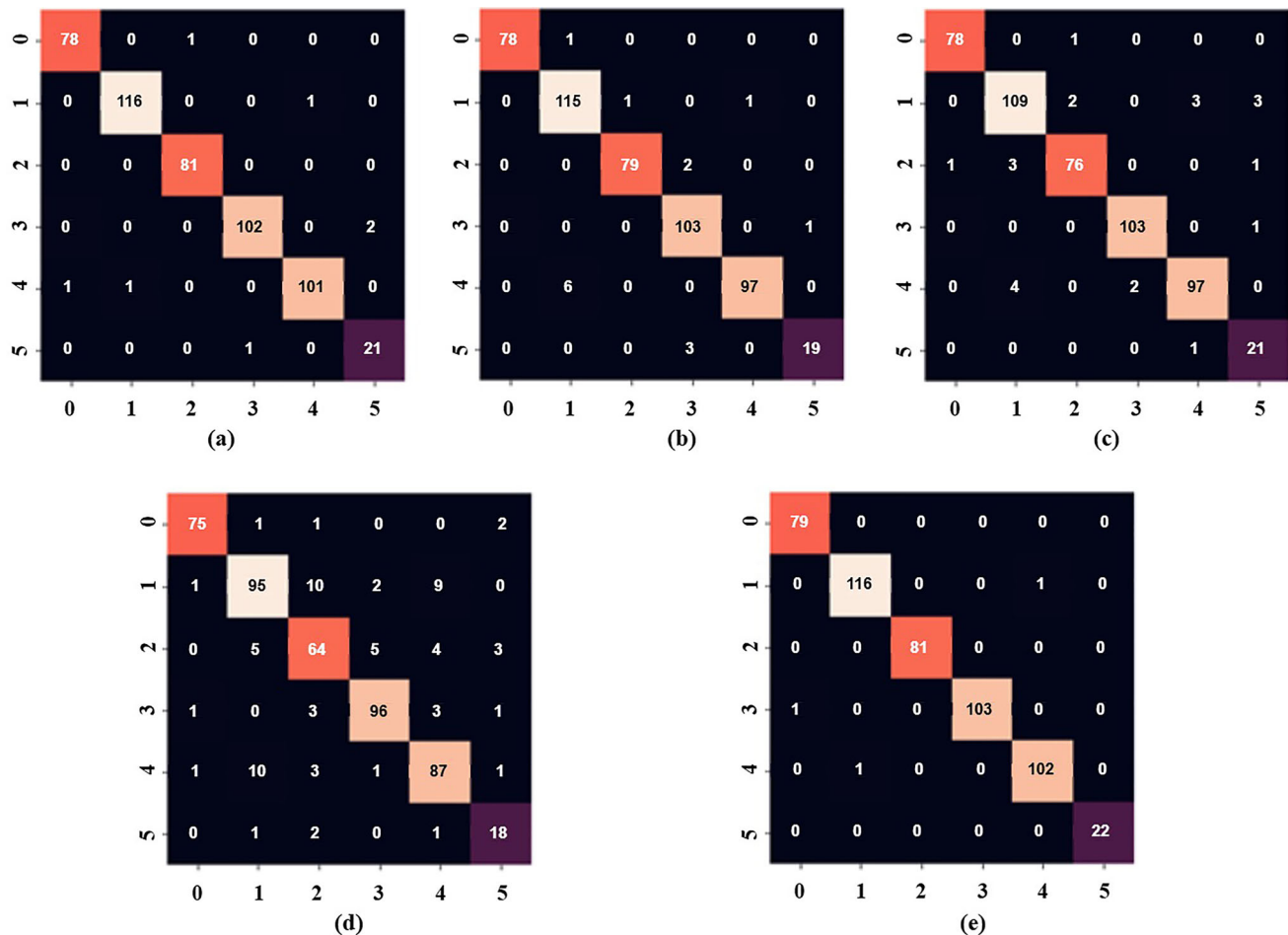
In the experimental study conducted on the TrashNet dataset using the developed models, the impact of features extracted from different layer levels on classification performance was analyzed. The findings are presented in Table 11. The results show that representations obtained from deeper layers significantly enhance model performance. In particular, the proposed model achieved the highest performance with 99.40% accuracy and 99.71% AUC by utilizing features from the deepest layers of EfficientNetB0 and InceptionV3. Model_1, which also uses deep-layer representations, achieved performance close to that of the proposed model (98.61% accuracy). Model_2, which uses mid-level representations, showed more limited success (97.03% accuracy), while Model_3, based on early-layer representations, reached a lower accuracy rate (95.65%). Model_4, which extracts features from very early layers close to the input, demonstrated the lowest performance (85.96% accuracy).

The confusion matrix results regarding the classification performance of the models developed for the TrashNet dataset are presented in Fig. 10. According to the findings,

Table 11 Test results of the developed models on the TrashNet dataset

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
Model_1	98.61	97.67	98.24	97.95	99.73	98.99
Model_2	97.03	97.06	95.69	96.32	99.38	97.53
Model_3	95.65	93.97	95.73	94.74	99.12	97.43
Model_4	85.96	84.34	85.62	84.92	97.13	91.38
Proposed Model	99.40	99.49	99.54	99.51	99.88	99.71

The bolded values indicate the best performance for each evaluation metric

**Fig. 10** Confusion matrix results of the models based on test images from the TrashNet dataset: **a** Model_1, **b** Model_2, **c** Model_3, **d** Model_4, and **e** Proposed Model

the proposed model correctly classified all samples in the 0th, 2nd, and 5th classes, while only one misclassification occurred in each of the other classes. These results demonstrate that the proposed model exhibits superior performance compared to the other models in terms of overall accuracy.

The test results conducted on the Household_Garbage dataset are presented in Table 12. The findings are consistent with those observed in the TrashNet dataset. The proposed model, trained with features extracted from deep

layers, achieved the highest performance across all metrics with 99.87% accuracy and 99.90% AUC. Model_1 also demonstrated high performance with features extracted from deep layers (99.64% accuracy), but results obtained from mid (Model_2) and early layers (Model_3 and Model_4) showed a gradual decline. In particular, the low accuracy and AUC values of Model_4 once again prove that early-stage features are insufficient for classification. These results clearly highlight the critical role of deep-layer representations in classification success, especially for complex datasets.

Table 12 Test results of the developed models on the Household_Garbage dataset

Method	Acc. (%)	Pre. (%)	Re. (%)	F1. (%)	Spe. (%)	AUC (%)
Model_1	99.64	99.41	99.48	99.44	99.97	99.72
Model_2	98.83	98.28	98.35	98.31	99.90	98.13
Model_3	96.87	95.69	95.78	95.71	99.72	97.75
Model_4	85.27	82.26	81.11	81.61	98.60	89.85
Proposed Model	99.87	99.77	99.81	99.79	99.99	99.90

The bolded values indicate the best performance for each evaluation metric

Figure 11 presents the confusion matrix results of the developed models for the test set of the Household_Garbage dataset, which contains 3103 images. Upon examining the results, it is observed that the proposed model correctly predicted all images in the 0th, 2nd, 3rd, 4th, 5th, 6th, 7th, 9th, and 10th classes. However, the proposed model misclassified one image in both the 1st and 8th classes, and only two images in the 11th class.

The experimental studies conducted on the TrashNet and Household_Garbage datasets clearly demonstrate that the features extracted from different layer levels have a direct impact on classification performance. In both datasets, the proposed model, built with representations extracted from deep layers, achieved the highest success across all performance metrics. Models trained with features extracted from deep layers showed significantly higher accuracy, precision, recall, specificity, and AUC values, while models built with representations from mid and early layer levels exhibited a gradual decrease in these values. These results highlight that deep feature representations play a critical role in improving model performance, particularly in complex classification problems, and emphasize the strategic importance of layer selection in classifier design.

8 Discussion

Improper management of solid waste is a serious problem that leads to global environmental pollution. With the increase in disposable products, the amount of waste is also increasing rapidly. Increasing consumption habits and living standards increase the amount of household waste and lead to the problem of "garbage siege" in many cities. Although waste classification is considered an important strategy for increasing environmental protection and resource efficiency, it cannot reach the desired level in practice due to a lack of public awareness and separation difficulties.

Some of the studies in the literature using TrashNet and different datasets on the classification of waste types are presented in Table 13. According to the findings presented in Table 13, the results obtained by the proposed model, especially in the Household_Garbage dataset, are extremely

remarkable. It is seen that the model achieves the highest performance in the literature with 99.87% Acc, 99.77% Pre, 99.81% Re, 99.79% F1, 99.99% Spe, and 99.90% AUC. This proves that the model is superior not only in one metric but in many important metrics and that its performance is balanced in different aspects.

While other studies, except for Hossen et al. [62], use only the accuracy metric, the evaluation of the proposed model with various metrics reveals the multifaceted performance of the model more comprehensively. The inclusion of metrics such as precision, recall, F1-score, and AUC allows us to understand the performance of the model more deeply. It is stated that the use of these metrics is critical for understanding how well the classification systems balance false positive and false negative results, as well as the accuracy of the classification systems, especially in real-world applications.

When the findings obtained with the TrashNet dataset are examined, it is seen that the proposed model exhibits the second highest performance in the literature after Mao et al. [38] with an accuracy rate of 99.40%. Mao et al.'s 99.6% accuracy obtained with the data augmentation method indicates that the model increases its learning capacity by artificially adding new images to the dataset. However, Mao et al. achieved only 94.02% accuracy without using data augmentation. The fact that our proposed model shows 99.40% accuracy without using data augmentation reveals that it provides much better results. This shows that the proposed model has a superior classification ability even with raw data.

When examining various studies in the field of waste classification in the literature, it is evident that deep learning-based architectures have made significant contributions in terms of accuracy and efficiency. The RecycleNet model developed by Bircanoğlu et al. [31], which employs a transfer learning approach, achieved a notable success with 95% accuracy on the TrashNet dataset. However, the model's high number of parameters and slow prediction time represent significant disadvantages, especially for real-time applications. Similarly, the M-b Xception model proposed by Shi et al. [34], instead of traditionally deepening networks, increased their width to achieve a classification performance of 94.34%. However, this success is largely based on customizations made with limited data, which can lead to higher

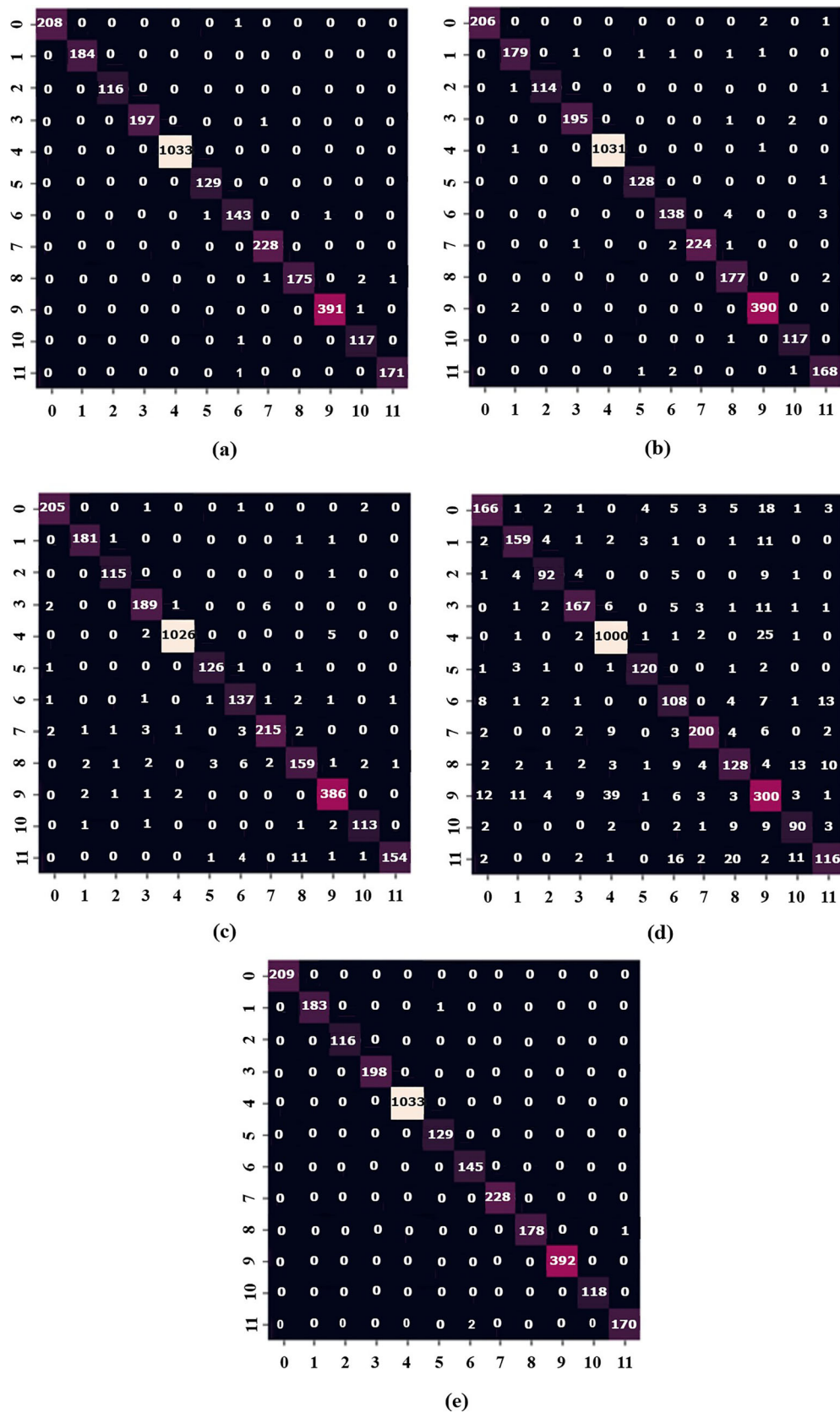


Fig. 11 Confusion matrix results of the models based on test images from the Household_Garbage dataset: (a) Model_1, **b** Model_2, c Model_3, **d** Model_4, and **e** Proposed Model

Table 13 Performance of some studies reviewed in the literature on waste image classification

Study	Dataset	Method	Acc	Pre	Re	F1	Spe	AUC
Bircanoğlu et al. [31], 2018	TrashNet	DenseNet121	95	–	–	–	–	–
Shi et al. [34], 2020	TrashNet	M-b Xception	94.34	–	–	–	–	–
Yang and Thung [39], 2016	TrashNet	SVM	63	–	–	–	–	–
Vo et al. [41], 2019	TrashNet	DNN-TC	94	–	–	–	–	–
Adedeji and Wang [35], 2019	TrashNet	ResNet-50 + SVM	87	–	–	–	–	–
Mao et al. [38], 2021	TrashNet	Genetic algorithm, DenseNet121, data augmentation	99.6	–	–	–	–	–
Mao et al. [38], 2021	TrashNet	Genetic algorithm, DenseNet121, no data augmentation	94.02	–	–	–	–	–
Huang et al. [61], 2020	TrashNet	Combined model, Transfer learning	96.5	–	–	–	–	–
Hossen et al. [62], 2024	GIGO	GCDN-Net	95.77	95.78	95.77	95.77	95.54	–
Li et al. [63], 2023	Private	GScbamKL-Net	81.26	–	–	–	–	–
Fue et al. [64], 2021	Huawei Garbage Classification Challenge Cup dataset	GNet	92.62	–	–	–	–	–
Kang et al. [65], 2020	Garbage	ResNet-34	89.96	–	–	–	–	–
Gupta et al. [66], 2022	TrashNet, Waste classification data, Drinking waste classification	InceptionNetV3	96.23	–	–	–	–	–
Nowakowski and Pamua [36], 2020	e-waste	CNN, R-CNN	%96.7	–	–	–	–	–
Proposed Model	TrashNet	Multi-Layer Feature Learning	99.40	99.49	99.54	99.51	99.88	99.71
Proposed Model	Household_Garbage	Multi-Layer Feature Learning	99.87	99.77	99.81	99.79	99.99	99.90

The bolded values indicate the best performance for each evaluation metric

resource consumption. The DNN-TC model developed by Vo et al. [41] also showed good performance with 94% accuracy on the TrashNet dataset, but the proposed model achieved a significant superiority with 99.40% accuracy compared to this structure.

On the other hand, the combined VGG19, DenseNet169, and NASNetLarge architectures by Huang et al. [61] produced successful results with accuracy rates of 96.5% and 94%, but this multi-architecture structure imposed a significant computational burden during both training and inference. The GCDN-Net architecture presented by Hossen et al. [62], which provides strong feature extraction through the deep integration of DenseNet201 and InceptionV3, also incurs high computational costs due to its 264-layer deep structure. More novel approaches like the GScbamKL-Net

model [63], which utilizes attention mechanisms (CBAM), LeakyReLU activation, and label smoothing, could only achieve 81.26% accuracy due to their complex structure, and have limited applicability on hardware with lower capabilities. Solutions like GNet, developed by Fue et al. [64], which can be integrated into embedded systems, have provided significant contributions to portability on low-cost hardware. However, it remains limited in classification accuracy with 92.62% accuracy and a decision time of 0.63 s. While the GNet model seems advantageous for embedded applications, the model proposed in this study distinguishes itself from existing models in the literature with its simple architecture, parametric efficiency, and high classification performance, offering superior accuracy and overall system performance.

In contrast, the model proposed in this study, which combines the EfficientNetB0 and InceptionV3 architectures with the HyperColumn technique, offers significant advantages in terms of both architectural efficiency and classification performance compared to previous models. This structure, which achieves 99.40% accuracy on the TrashNet dataset and 99.87% accuracy on the Household_Garbage dataset, not only provides high accuracy but also effectively manages class imbalances with the XGBClassifier. The proposed model, thanks to its structure that extracts more information with fewer parameters, presents a highly generalizable and computationally efficient classification process.

However, while the proposed model stands out with high classification accuracy and architectural efficiency, it does have several limitations in specific application scenarios. The model contains a considerable number of parameters and requires substantial computational power, due to the reliance on deep and complex architectures such as EfficientNetB0 and InceptionV3. This limits the direct applicability of the model, especially in resource-constrained environments such as embedded systems, mobile devices, or IoT platforms. Furthermore, the high dimensionality of features extracted from deep layers increases memory usage, leading to longer training and inference times. Such an architectural structure may cause delays in real-time applications and could also create disadvantages in terms of energy consumption. The training time of the model, due to architectural complexity, may extend, and hyperparameter optimization and retraining processes may require significant engineering effort. These factors could complicate the sustainability and maintenance processes of the model. While the proposed model stands out as a powerful classifier on systems with high hardware capacities, it requires further optimization for applicability in resource-constrained environments.

9 Future works

Although the performance of the current model in terms of classification accuracy and generalizability at the dataset level is quite satisfactory, some improvements are needed at the application level. In particular, simplifying the model structurally to adapt it to resource-constrained environments (e.g., embedded systems, mobile applications, edge computing devices) is an important direction for future work. To this end, techniques such as model pruning, knowledge distillation, and quantization are targeted to lighten the model. This would reduce memory usage and shorten inference time, thus making it more suitable for real-time applications. Additionally, hybrid structures will be explored by experimenting with alternative models that have lower parameter counts (e.g., MobileNetV3, ShuffleNet, SqueezeNet), aiming to develop

more efficient solutions without compromising classification performance.

Moreover, testing the proposed model on larger and more balanced datasets covering different types of waste will enhance the model's generalizability. Particularly, addressing class imbalance using more advanced techniques (e.g., focal loss, class-balanced sampling) could further improve the model's performance. Additionally, efforts on hyperparameter optimization and network design will facilitate the development of more efficient and higher-performing models.

Another key area of research is the integration of the proposed model with components like sensor data, user interaction, and robotic control. Such integration could enable the creation of a more comprehensive waste sorting system for real-world applications. Integrated solutions used in smart cities and waste management systems could make waste collection processes more efficient and aim to enhance environmental sustainability.

Future work should not only focus on classification performance but also adopt a multidisciplinary approach that includes factors like system scalability, sustainability, and user experience. In this context, further research and optimization are required to improve and make waste management and classification systems more efficient and applicable on a larger scale. These efforts aim to ensure that the model works more efficiently and effectively under real-world conditions and make significant contributions to environmental protection, which is one of our key objectives.

10 Conclusion

In this study, a deep learning-based model is proposed for waste classification using the Household_Garbage and TrashNet datasets. The model employs an ensemble learning approach that integrates EfficientNetB0, InceptionV3, and the HyperColumn technique. By combining these architectures with HyperColumn, the model effectively captures multi-resolution features, enhancing its ability to learn both general and fine-grained patterns. The fusion of feature maps obtained from EfficientNetB0 and InceptionV3 using HyperColumn results in a richer feature representation by leveraging information from multiple layers. Consequently, the model processes both high-level and detailed features more efficiently.

For the classification task, the XGBClassifier is utilized within the ensemble framework, optimizing the final classification performance and improving overall accuracy. The results demonstrate that the proposed model outperforms other approaches on both the Household_Garbage and TrashNet datasets. Specifically, on the Household_Garbage

dataset, the model achieved 99.87% accuracy, 99.77% precision, 99.81% recall, 99.79% F1-score, 99.99% specificity, and 99.90% AUC. These metrics confirm that the model not only delivers high accuracy but also maintains a balanced performance by minimizing false positives and false negatives. Notably, the 99.99% specificity value highlights the model's ability to nearly eliminate false positive predictions. Similarly, the model demonstrated strong performance on the TrashNet dataset, achieving 99.40% accuracy, 99.49% precision, 99.54% recall, 99.51% F1-score, 99.88% specificity, and 99.71% AUC. These consistent results across different datasets validate the robustness and generalizability of the proposed approach.

Additionally, the classification performance of the proposed model has been evaluated in detail on both the TrashNet and Household_Garbage datasets. When examining the confusion matrix results for the TrashNet dataset, it is observed that the model achieved 100% accuracy in the Cardboard (0), Metal (2), and Trash (5) classes. For other classes, small-scale misclassifications were observed between similar classes; for example, an example from the Glass (1) class was misclassified as Plastic (4), an example from the Paper (3) class was misclassified as Cardboard (0), and an example from the Plastic (4) class was misclassified as Glass (1). The test results on the Household_Garbage dataset also support the high accuracy potential of the proposed architecture. The model accurately predicted all examples in the battery (0), brown-glass (2), cardboard (3), clothes (4), green-glass (5), metal (6), paper (7), shoes (9), and trash (10) classes. Additionally, one example from the biological (1) class was misclassified as green-glass (5), two examples from the white-glass (11) class were misclassified as metal (6), and one example from the plastic (8) class was misclassified. These results demonstrate that the proposed model has high discriminative power across different classes and exhibits strong generalization ability in both datasets.

The experimental analyses conducted on the TrashNet and Household_Garbage datasets have also revealed that the features extracted from the different convolutional layers of the EfficientNetB0 and InceptionV3 architectures have a decisive impact on classification performance. Specifically, the proposed model, which uses representations extracted from deeper layers, has shown superior results across all performance metrics such as accuracy, precision, specificity, sensitivity, and AUC. In contrast, models using representations from intermediate and early layers have shown a gradual decrease in these metrics. These findings highlight that deep feature representations are an important factor in enhancing success in complex visual classification tasks and that selecting the right layers in the classifier architecture is a strategic decision.

Overall, the findings demonstrate that the ensemble learning strategy, which integrates EfficientNetB0, InceptionV3,

and the HyperColumn technique, offers an innovative and highly effective solution for waste classification. The model's strong performance across both datasets highlights the potential of combining multiple architectures to improve classification accuracy and reliability in waste image analysis.

Author contribution G.C. was responsible for the conception and design of the study, data collection, analysis, and interpretation, as well as writing and revising the manuscript. All aspects of the work were carried out independently by the author.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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